

NON-MF GROUPS

SAUERS

We prove that not every countable group is MF in the operator norm sense. Let $L \leq G$ and $K \trianglelefteq G$ have property (T). If K is contained in the one-sided compression defect of L , then K lies in the MF radical of G . For $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$, this defect equals H . The group H is finitely generated, simple, has property (T), and has trivial image under every homomorphism to an MF group. Consequently, $C_r^*(H)$ is a separable stably finite C^* -algebra that is not MF. We also construct a finitely presented torsion-free property-(T) group with full MF radical; every nontrivial quotient of it again has full MF radical.

Draft. This manuscript is under revision.

1. INTRODUCTION

Not every countable group is MF. The counterexample is

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

Khanh–Thanh identify H with the unit group of the binary Leavitt algebra [K–T, Proposition 4.2 and Corollary 4.4], which OpenAI used to construct the first nonsofic group [OAI, Chapter 3].

A countable group G is *MF* [CDE] if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

We use this operator norm convention throughout. The strong convergence convention additionally requires the matrix models to recover the norms of the left regular representation [GKM, Sch].

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here $\bigoplus_n M_{d_n}(\mathbb{C})$ is the ideal of sequences (x_n) for which $\|x_n\| \rightarrow 0$.

A separable C^* -algebra is MF if it embeds in a norm matrix corona. For a countable group G , the algebra $C_r^*(G)$ is separable and its faithful canonical

2020 *Mathematics Subject Classification*. Primary 46L05; Secondary 20C07, 16S50, 20G35.

Key words and phrases. MF groups, norm matrix coronas, MF radicals, property (T), asymptotic representations, Leavitt algebra.

trace makes it stably finite. The group H above gives a non-MF algebra of this form.

For a countable group G , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism $G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. By Proposition 2.1, G is MF exactly when $\text{Rad}_{\text{MF}}(G) = 1$, while $\text{Rad}_{\text{MF}}(G) = G$ implies that every homomorphism from G to an MF group is trivial.

For a subgroup $L \leq G$, put

$$\text{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression-centralizer defect* by

$$(1) \quad \mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \text{Comp}_G(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

Theorem A (one-sided compression criterion). *Let G be countable and let $L \leq G$ have property (T). If $K \trianglelefteq G$ has property (T) and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) and $\mathfrak{D}_G(L) = G$, then $\text{Rad}_{\text{MF}}(G) = G$.

Every operator norm asymptotic representation is asymptotically trivial on $\mathfrak{D}_G(L)$ in normalized Hilbert–Schmidt norm:

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Conjugation by the compressor moves the Kazhdan projection of L below itself, and stable finiteness of the corona forces equality. For a corona representation, the complement of the Kazhdan projection is a G -invariant corner. The Kazhdan inequality forces this corner to vanish when $K \leq \mathfrak{D}_G(L)$.

Many Kazhdan groups are residually finite and so MF.

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(2) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group). *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{12}(R)$. Then H is finitely generated, nontrivial, simple, has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

Every homomorphism from H to an MF group is trivial. So H is non-MF, and $C_r^(H)$ is separable and stably finite but not MF.*

Theorem C (a torsion-free group with full MF radical). *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property (T) and*

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

Every nontrivial quotient of Q also equals its own MF radical; in particular, no nontrivial quotient of Q is MF.

The construction combines Fournier-Facio's torsion-free property-(T) group [FF, §2] with Hull's small cancellation theorem [Hu] so that the compression defect normally generates Q .

Relation to prior work. Blackadar and Kirchberg developed the notions of MF and NF algebras and proved that a separable C^* -algebra is NF exactly when it is nuclear and MF [BK]. Carrión–Dadarlat–Eckhardt introduced the group property used here [CDE]. The negative solution of the Connes embedding problem [JNVWY] implies the existence of separable stably finite C^* -algebras that are not MF [GH, Proposition 6.1 and Remark 6.2]; Theorem B gives one among reduced group C^* -algebras. Nonapproximability was known for the unnormalized Schatten p -norms when $1 < p < \infty$ [DGLT, LO], and Bachner–Dogon–Lubotzky settled $p = 1$ [BDL]. The case $p = \infty$ remained open. Their criterion reduces the operator norm case to normalized Hilbert–Schmidt stability [BDL, Proposition 1.5]. Under the compression relation, conjugation preserves the normalized Hilbert–Schmidt asymptotic commutant.

Leavitt introduced the algebras $L_K(1, n)$ [Lev]. The binary Leavitt algebra is purely infinite simple [AA], is an exchange ring [Ara], and has center $\mathbb{F}_2$ [AC]. Preusser's sandwich theorem then implies that H is simple [Pre]. Property (T) follows from Ershov–Jaikin–Zapirain [EJZ, Theorem 1.1].

2. ONE-SIDED COMPRESSION AND THE MF RADICAL

2.1. The MF radical. The image of a corona homomorphism from G is a countable subgroup of a corona unitary group, so MF; conversely, every countable MF group embeds in the unitary group of a norm matrix corona. Thus $\text{Rad}_{\text{MF}}(G)$ is also the intersection of the kernels of all homomorphisms from G to MF groups.

Proposition 2.1 (basic properties of the MF radical). *For every countable group G , the subgroup $\text{Rad}_{\text{MF}}(G)$ is fully invariant, the quotient $G/\text{Rad}_{\text{MF}}(G)$ is MF, and G is MF if and only if $\text{Rad}_{\text{MF}}(G)$ is trivial.*

Proof. An intersection of kernels is normal. If $\alpha: G \rightarrow G$ is an endomorphism and $x \in \text{Rad}_{\text{MF}}(G)$, then every corona homomorphism π satisfies $\pi(\alpha(x)) = (\pi \circ \alpha)(x) = 1$. Thus the radical is fully invariant.

Put $R = \text{Rad}_{\text{MF}}(G)$. If $1 \neq x \in G/R$ and g represents x , then $g \notin R$, so a homomorphism from G to an MF group separates g from 1. It kills R and induces a homomorphism separating x . So G/R is residually MF, and MF by Korchagin [Kor, Proposition 6]; this also covers the trivial quotient.

Finally, a corona embedding gives $\text{Rad}_{\text{MF}}(G) = 1$ when G is MF, and the quotient result gives the converse. $\square$

2.2. Kazhdan transport in normalized Hilbert–Schmidt norm. For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \rightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G . Define

$$(3) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where V ranges over all operator norm asymptotic representations of G .

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \rightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\text{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.2. *For every sequence of positive integers (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Write A for the displayed corona. Suppose that $v^*v = 1$ in A , and let (x_n) be a bounded lift of v . Then $\|x_n^*x_n - 1\| \rightarrow 0$. For all sufficiently large n , the matrix $x_n^*x_n$ is invertible, and

$$u_n = x_n(x_n^*x_n)^{-1/2}$$

is unitary. Moreover, $\|u_n - x_n\| \rightarrow 0$; after assigning arbitrary unitary values to the finitely many remaining coordinates, (u_n) is a unitary lift of v . So $vv^* = 1$, and A is finite. For every $k \geq 1$, apply this construction with matrix sizes km_n to see that $\prod_n M_{km_n}(\mathbb{C}) / \bigoplus_n M_{km_n}(\mathbb{C})$ is finite. This corona is canonically isomorphic to $M_k(A)$, so A is stably finite. Finally, let $p \leq q$ be equivalent projections in a stably finite algebra B , and choose a partial

isometry w with $w^*w = q$ and $ww^* = p$. Then $w \in qBq$ is an isometry in the finite corner qBq , so $p = ww^* = q$. $\square$

Lemma 2.3 (one-sided order for the Kazhdan projection). *Let L have property (T), let B be a unital C^* -algebra, and let $\pi: L \rightarrow \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C_{\max}^*(L) \rightarrow B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\text{Fix } \pi(L)$. If ξ is L -fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. So $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B . $\square$

Theorem 2.4 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the conjugation actions are multiplicative modulo c_0 because the multiplicative defects of (V_n) tend to zero in operator norm. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2 . By the preceding estimate, $g \mapsto \tilde{\sigma}(g)$ is a homomorphism $G \rightarrow \mathcal{U}(\mathcal{B})$. So its restriction to L extends to a $*$ -homomorphism $C_{\max}^*(L) \rightarrow \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.3, and so $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 2.2 then

gives $U^*PU = P$: U commutes with P , and both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\text{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. $\square$

Corollary 2.5. *Let $L \leq G$ have property (T), let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G) \quad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 2.4, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \rightarrow 0.$$

So

$$\|V_n([ucu^{-1}, \ell]) - 1\|_2 \rightarrow 0 \quad (\ell \in L).$$

Intersecting over (V_n) proves the claim. $\square$

2.3. Normal Kazhdan subgroups and the MF radical.

Lemma 2.6 (central corona corners). *Let $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \text{rank}(q_n)$, unitary identifications $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n: G \rightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_n W_n(g) J_n^)$ is the coordinate restriction of $q\rho(g)$.*

Proof. Lift q to projections q_n , and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \rightarrow 0 \quad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. For each fixed g , let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\|\widetilde{W}_n(g) - q_n U_n(g) q_n\| \rightarrow 0.$$

For $g = 1$, the compression is already the identity q_n of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain and relabel

those coordinates. Put $r_n = \text{rank}(q_n)$, choose a unitary $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by J_n preserves operator norms and multiplicative defects. Thus (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. $\square$

Theorem 2.7 (normal Kazhdan radical theorem). *Let G be countable and let $K \trianglelefteq G$ have property (T) with $K \leq R_{\infty \rightarrow 2}(G)$. Then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K . Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ . The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. It lies in $R_{\infty \rightarrow 2}(\Theta(G))$, because every operator norm asymptotic representation of $\Theta(G)$ pulls back along $G \rightarrow \Theta(G)$ to one of G .

Let $p_K \in C_{\max}^*(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K -fixed vectors. Normality of K gives

$$\Theta(g)p\Theta(g)^* = p \quad (g \in G).$$

The two ranges agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K , so every element of K has trivial corona image. Lemma 2.6 gives positive integers r_n and an operator norm asymptotic representation $W_n: G \rightarrow \mathcal{U}(r_n)$. Let $\hat{\Theta}: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$ be its corona homomorphism. Under the coordinatewise corner identifications, $\hat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. Represent $\mathcal{Q}_{\mathbf{r}}$ faithfully. The Kazhdan projection of K under $\hat{\Theta}$ is zero, so $\hat{\Theta}|_K$ has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\hat{\Theta}(s) - 1)^* (\hat{\Theta}(s) - 1)$$

the inequality

$$b \geq \frac{\kappa^2}{|S|} 1.$$

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - I_{r_n})^* (W_n(s) - I_{r_n})$$

represent b . Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ then stays a positive Hilbert–Schmidt distance from I_{r_n} . This contradicts $s_0 \in K \leq R_{\infty \rightarrow 2}(G)$, established after passing to $\Theta(G)$. Thus every corona homomorphism kills K . $\square$

Proof of Theorem A. For every $u \in \text{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 2.5 gives

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G).$$

So normality of $R_{\infty \rightarrow 2}(G)$ gives

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Since $K \leq \mathfrak{D}_G(L)$, Theorem 2.7 now gives $K \leq \text{Rad}_{\text{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\text{Rad}_{\text{MF}}(G) = G$. $\square$

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.4.

3. THE BINARY LEAVITT GROUP

The Leavitt relations give $\tau \in H$ with $\tau L \tau^{-1} \leq L$ for a copy L of $\text{EL}_3(R)$ and a nontrivial $d \in \mathfrak{D}_H(L)$. Simplicity of H then gives $\mathfrak{D}_H(L) = H$.

3.1. The self-compression. Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0 t_0, \quad q = s_1 t_1.$$

The relations (2) give

$$(4) \quad p + q = 1, \quad t_1 q s_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\text{GL}_3(R)$ with the subgroup of $\text{GL}_{12}(R)$ consisting of the matrices $\text{diag}(A, I_9)$. Define a map $\Psi: M_3(R) \rightarrow M_3(R)$ by

$$(5) \quad \Psi(A) = qI_3 + s_0 A t_0.$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q^2 = q$, $q s_0 = t_0 q = 0$, and $t_0 s_0 = 1$, all consequences of (2), show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0 I_3 t_0 = qI_3 + pI_3 = (p + q)I_3 = I_3,$$

so Ψ preserves the identity. Finally, $t_0\Psi(A)s_0 = A$, so Ψ is injective and restricts to an injective group homomorphism $\mathrm{GL}_3(R) \rightarrow \mathrm{GL}_3(R)$.

To realize Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \quad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $Y = X^{-1}$. Put

$$(6) \quad \tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R).$$

A direct block multiplication gives

$$(7) \quad \tau \mathrm{diag}(A, I_9) \tau^{-1} = \mathrm{diag}(\Psi(A), I_9).$$

Lemma 3.1. *The matrix τ defined in (6) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. For $B \in M_6(R)$, the block matrices $\begin{pmatrix} I & B \\ 0 & I \end{pmatrix}$ and $\begin{pmatrix} I & 0 \\ B & I \end{pmatrix}$ lie in $\mathrm{EL}_{12}(R)$: they are the products of the elementary matrices $e_{r,6+s}(b_{rs})$, respectively $e_{6+r,s}(b_{rs})$, and no cross terms arise because distinct off-diagonal matrix units have product zero. Whitehead's identity

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$$

with $I = I_6$ then expresses τ as a product of six matrices of $\mathrm{EL}_{12}(R)$. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \leq H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property (T), and $\tau \in H$ satisfies*

$$(8) \quad \tau L \tau^{-1} \leq L.$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. So the theorem of Ershov and Jaikin-Zapirain gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$.

For $i \neq j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L . These matrices generate L , so (8) follows. $\square$

3.2. Simplicity and the full MF radical.

Proposition 3.3. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], and so is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \mathrm{EL}_{12}(R)$, the subgroup $N \leq \mathrm{GL}_{12}(R)$

is normalized by $\text{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \leq R$ such that

$$\text{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of R gives $I = 0$ or $I = R$. If $I = R$, then $\text{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \leq C_{12}(R, 0) = Z(\text{GL}_{12}(R))$. Fix $g \in N$. Since g centralizes $e_{ij}(1)$ for every $i \neq j$, comparison of entries in $gE_{ij} = E_{ij}g$ shows that $g = \lambda I_{12}$ for some $\lambda \in R^\times$. Centralizing $e_{ij}(a)$ now gives $\lambda a = a\lambda$ for every $a \in R$, and so $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $g = 1$. Thus $N = 1$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 3.4. *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. So the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L , so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$(9) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ then gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Since H is simple, the nontrivial element d normally generates H . $\square$

Proof of Theorem B. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. The identities $e_{ij}(a+b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ show inductively that Y generates all elementary matrices over R ; the commutator identity always has a third index because $12 \geq 3$. Thus $H = \langle Y \rangle$ is finitely generated, and Proposition 3.2 gives property (T).

The containment (8), the centrality of c relative to L , and Proposition 3.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 3.4 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\text{Rad}_{\text{MF}}(H) = H$. Proposition 3.3 shows that H is nontrivial and simple.

The equality $\text{Rad}_{\text{MF}}(H) = H$ implies that every homomorphism from H to an MF group is trivial, and so that H is not MF.

$C_r^*(H)$ is separable, and its faithful canonical trace makes it stably finite. If an MF embedding $e: C_r^*(H) \rightarrow \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1-p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, contrary to the preceding conclusion. $\square$

4. A TORSION-FREE GROUP WITH FULL MF RADICAL

For an acylindrically hyperbolic group G , choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. We use Hull's small cancellation theorem [Hu, Theorem 7.1] as follows. Given a finite set $F \subseteq G$, choose a ball in $\Gamma(G, A)$ containing F and apply Hull's injectivity-radius formulation. This gives the finite-set version recorded below.

Theorem 4.1 (Hull's small cancellation theorem). *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A , and let $t_1, \dots, t_m \in G$ and a finite subset $F \subseteq G$ be given. Then there is a surjective homomorphism $q: G \rightarrow Q$ such that Q is acylindrically hyperbolic, $q|_F$ is injective, $q(t_i) \in q(N)$ for all i , and every element of finite order in Q is the image of an element of the same order in G .*

Separately, inspection of Hull's construction, applied successively to $t_1, \dots, t_m$, shows that $\ker q$ is normally generated by m elements.

Lemma 4.2 (saturation). *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $F \subseteq G$ be finite. Then there is a surjective homomorphism $q: G \rightarrow Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q|_F$ is injective, and $q(N) = Q$.*

Proof. Because $N \trianglelefteq G$, one has $N \cap gNg^{-1} = N$ for every $g \in G$; this group is infinite because G is torsion-free and $N \neq 1$. Thus N acts non-elementarily on $\Gamma(G, A)$ by Osin [Os, Lemma 7.1]. Torsion-freeness makes N suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 4.1 to N_0 with $t_1, \dots, t_m$ a finite generating set of G and with the set F . Then

$$Q = \langle q(t_1), \dots, q(t_m) \rangle \leq q(N_0) \leq q(N) \leq Q,$$

so $Q = q(N_0) = \langle q(h_1), q(h_2) \rangle$ and $q(N) = Q$. The quotient is torsion-free by Theorem 4.1 and finitely presented because $\ker q$ is normally generated by m elements. $\square$

Following Fournier-Facio [FF, §2], let U be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group [Chi]; let S be a finitely presented infinite simple torsion-free group [HL]; and let H_0 be a torsion-free hyperbolic group with property (T), obtained from the density model at a parameter between $1/3$ and $1/2$ [KK, Ol]. Small cancellation over the relatively hyperbolic pair $(U * H_0, U)$ embeds U in a finitely presented torsion-free quotient P of H_0 [FF2, Proposition 2.3]; torsion-freeness is preserved by Osin's embedding theorem [OsSC, Theorem 2.4(5)]. Consequently, P has property (T). By universality, P contains a subgroup $P_1 \times P_2 \times S$ with $P_i \cong P$. Let E be the double HNN extension

of P with stable letters u_1, u_2 and $u_i P u_i^{-1} = P_i$; it is finitely presented and torsion-free, and its Bass–Serre action makes it acylindrically hyperbolic by the tree criterion of Minasyan–Osin [MO]. Hull’s common quotient theorem [Hu, Corollary 7.4] applied to E and H_0 gives a surjection $\pi: E \rightarrow G_0$ onto a finitely presented torsion-free group with property (T), injective on a prescribed finite subset of E ; since E is finitely generated, the proof of that corollary produces G_0 acylindrically hyperbolic. We run the construction with the prescribed finite subset containing an element $s \neq 1$ of S . Then $\pi(S) \neq 1$, so $\pi|_S$ is injective because S is simple.

Put $\Gamma = \pi(P)$, $t = \pi(u_1)$, and $J = t^{-1}\pi(S)t$. Since S commutes with $P_1 = u_1 P u_1^{-1}$, the subgroup J centralizes Γ , and $t\Gamma t^{-1} = \pi(P_1) \leq \Gamma$, so $t \in \text{Comp}_{G_0}(\Gamma)$. Moreover $tJt^{-1} = \pi(S) \leq \Gamma$, and Γ has property (T) as a quotient of P .

Lemma 4.3. *Let $\rho: G_0 \rightarrow L$ be a surjective homomorphism with $\rho(\pi(S)) \neq 1$. Then $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$.*

Proof. Write $\Gamma_L = \rho(\Gamma)$, $t_L = \rho(t)$, and $J_L = \rho(J)$. Then $t_L \Gamma_L t_L^{-1} \leq \Gamma_L$, J_L centralizes Γ_L , and $t_L J_L t_L^{-1} = \rho(\pi(S)) \leq \Gamma_L$. For $c \in J_L$ and $\ell \in \Gamma_L$, the commutator $[t_L c t_L^{-1}, \ell]$ lies in $\mathfrak{D}_L(\Gamma_L)$ by (1). As c ranges over J_L , the element $t_L c t_L^{-1}$ ranges over $\rho(\pi(S))$, and ℓ may be taken in $\rho(\pi(S)) \leq \Gamma_L$; so the commutator subgroup of $\rho(\pi(S))$ lies in $\mathfrak{D}_L(\Gamma_L)$. Since $\rho(\pi(S)) \neq 1$ and S is simple, $\rho(\pi(S)) \cong S$ is perfect, so $\rho(\pi(S)) \leq \mathfrak{D}_L(\Gamma_L)$. $\square$

Proof of Theorem C. Let N be the normal closure of $\pi(S)$ in G_0 . It is nontrivial, so Lemma 4.2 applied to G_0 , N , and $F = \{1, \pi(s)\}$ gives $q: G_0 \rightarrow Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q(N) = Q$, and $q(\pi(s)) \neq 1$. The group Q has property (T) as a quotient of G_0 .

Let $r: Q \rightarrow L$ be a surjection onto a nontrivial group and put $\rho = r \circ q$. Since $q(N) = Q$, the normal closure of $\rho(\pi(S))$ in L is $\rho(N) = L$; as $L \neq 1$, this forces $\rho(\pi(S)) \neq 1$. By Lemma 4.3, $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$, and since $\mathfrak{D}_L(\rho(\Gamma))$ is normal in L and contains a normal generating set of L , it equals L . The group L has property (T) as a quotient of Q , and $\rho(\Gamma)$ has property (T) as a quotient of Γ . Theorem A, applied to L with the subgroup $\rho(\Gamma)$ and $K = L$, gives $\text{Rad}_{\text{MF}}(L) = L$. Taking r the identity gives $\text{Rad}_{\text{MF}}(Q) = Q$, and a nontrivial group equal to its own MF radical is not MF. $\square$

ACKNOWLEDGMENTS

The author thanks Francesco Fournier-Facio for early advice and detailed comments on earlier drafts, and Alon Dogon for organizing the ensuing discussion and for comments on the surrounding literature. The author also thanks Tatiana Shulman for clarifying the terminology and historical context of the MF question, and Caleb Eckhardt for discussions of MF representations and matrix norms.

REFERENCES

- [AA] G. Abrams and G. Aranda Pino, *Purely infinite simple Leavitt path algebras*, J. Pure Appl. Algebra **207** (2006), no. 3, 553–563, [doi:10.1016/j.jpaa.2005.10.010](#).
- [AW] C. A. Akemann and M. E. Walter, *Unbounded negative definite functions*, Canad. J. Math. **33** (1981), no. 4, 862–871.
- [AC] G. Aranda Pino and K. Crow, *The center of a Leavitt path algebra*, Rev. Mat. Iberoam. **27** (2011), no. 2, 621–644, [doi:10.4171/RMI/649](#).
- [Ara] P. Ara, *The exchange property for purely infinite simple rings*, Proc. Amer. Math. Soc. **132** (2004), no. 9, 2543–2547, [doi:10.1090/S0002-9939-04-07369-1](#).
- [BDL] B. Bachner, A. Dogon, and A. Lubotzky, *On L^1 -approximation of groups*, J. Algebra **702** (2026), 235–243, [doi:10.1016/j.jalgebra.2026.04.041](#), [arXiv:2508.17392](#).
- [BK] B. Blackadar and E. Kirchberg, *Generalized inductive limits of finite-dimensional C^* -algebras*, Math. Ann. **307** (1997), no. 3, 343–380, [doi:10.1007/s002080050039](#).
- [CDE] J. R. Carrión, M. Dadarlat, and C. Eckhardt, *On groups with quasidiagonal C^* -algebras*, J. Funct. Anal. **265** (2013), no. 1, 135–152, [doi:10.1016/j.jfa.2013.04.004](#).
- [Chi] M. Chiodo, *On torsion in finitely presented groups*, Groups Complex. Cryptol. **6** (2014), no. 1, 1–8, [arXiv:1107.1489](#).
- [DGLT] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and nonapproximable groups*, Forum Math. Sigma **8** (2020), Paper No. e18, [doi:10.1017/fms.2020.5](#).
- [EJZ] M. Ershov and A. Jaikin-Zapirain, *Property (T) for noncommutative universal lattices*, Invent. Math. **179** (2010), 303–347, [doi:10.1007/s00222-009-0218-2](#).
- [FF] F. Fournier-Facio, *A torsion-free non-sofic group*, preprint, 2026, [arXiv:2608.02025](#).
- [FF2] F. Fournier-Facio, *Stability, approximable quotients, and higher property (T)*, preprint, 2025, [arXiv:2512.09180](#).
- [GKM] D. Gao, S. Kunnawalkam Elayavalli, A. Manzoor, and G. Patchell, *A new source of purely finite matricial fields*, preprint, 2026, [arXiv:2603.24502](#).
- [GH] I. Goldbring and B. Hart, *The universal theory of the hyperfinite II_1 factor is not computable*, Bull. Symbolic Logic **30** (2024), no. 2, 181–198, [doi:10.1017/bsl.2024.7](#), [arXiv:2006.05629](#).
- [Hu] M. Hull, *Small cancellation in acylindrically hyperbolic groups*, Groups Geom. Dyn. **10** (2016), no. 4, 1077–1119, [doi:10.4171/GGD/377](#).
- [HL] J. Hyde and Y. Lodha, *Finitely presented simple left-orderable groups in the landscape of Richard Thompson’s groups*, Ann. Sci. Éc. Norm. Supér. (4) **58** (2025), no. 2, 419–432, [arXiv:2302.04805](#).
- [JNVWY] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, $\text{MIP}^* = \text{RE}$, Commun. ACM **64** (2021), no. 11, 131–138, [doi:10.1145/3485628](#), [arXiv:2001.04383](#).
- [K–T] H. V. Khanh and V. H. Thanh, *Matrix generators for the unit groups of $L_K(1, d)$* , preprint, 2026, [arXiv:2607.10351](#).
- [Kor] A. I. Korchagin, *The MF-property for countable discrete groups*, Math. Notes **102** (2017), no. 2, 198–211, [doi:10.1134/S0001434617070227](#), [arXiv:1704.06906](#).
- [KK] M. Kotowski and M. Kotowski, *Random groups and property (T): Żuk’s theorem revisited*, J. Lond. Math. Soc. (2) **88** (2013), no. 2, 396–416, [arXiv:1106.2242](#).
- [Lev] W. G. Leavitt, *The module type of a ring*, Trans. Amer. Math. Soc. **103** (1962), 113–130, [doi:10.1090/S0002-9947-1962-0132764-X](#).
- [LO] A. Lubotzky and I. Oppenheim, *Non p -norm approximated groups*, J. Anal. Math. **141** (2020), no. 1, 305–321, [doi:10.1007/s11854-020-0119-2](#).
- [MO] A. Minasyan and D. Osin, *Acylindrical hyperbolicity of groups acting on trees*, Math. Ann. **362** (2015), no. 3–4, 1055–1105, [arXiv:1310.6289](#).

- [OAI] OpenAI, *Nonsofic groups exist*, in *Ten advances in mathematics and theoretical computer science*, Chapter 3, 78–95, 2026, <https://cdn.openai.com/pdf/ten-proofs-oai.pdf>.
- [Ol] Y. Ollivier, *Sharp phase transition theorems for hyperbolicity of random groups*, Geom. Funct. Anal. **14** (2004), no. 3, 595–679, [arXiv:math/0301187](https://arxiv.org/abs/math/0301187).
- [Os] D. Osin, *Acylindrically hyperbolic groups*, Trans. Amer. Math. Soc. **368** (2016), no. 2, 851–888, [doi:10.1090/tran/6343](https://doi.org/10.1090/tran/6343).
- [OsSC] D. Osin, *Small cancellations over relatively hyperbolic groups and embedding theorems*, Ann. of Math. (2) **172** (2010), no. 1, 1–39, [arXiv:math/0411039](https://arxiv.org/abs/math/0411039).
- [Pre] R. Preusser, *On general linear groups over exchange rings*, Linear and Multilinear Algebra **70** (2022), no. 4, 705–713, [doi:10.1080/03081087.2020.1743636](https://doi.org/10.1080/03081087.2020.1743636).
- [Sch] C. Schafhauser, *Finite dimensional approximations of certain amalgamated free products of groups*, preprint, 2023, [arXiv:2306.02498](https://arxiv.org/abs/2306.02498).